

Optimized UAV Service Network

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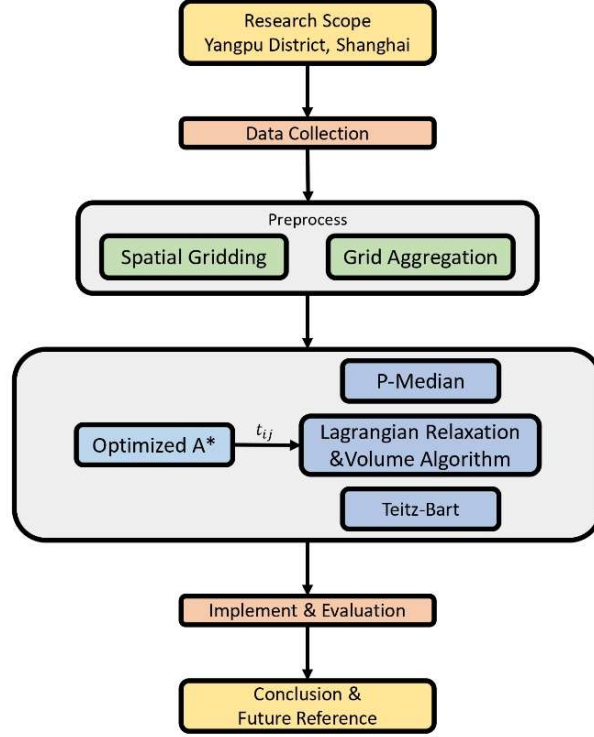
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Abstract. The booming low-altitude economy is now impelling urban development worldwide, with unmanned aerial vehicle service networks empowering city governance, emergency services and day-to-day deliveries. Nevertheless, high-rise building clusters, complex low-altitude airspace regulations and uneven layout of UAV vertiports are making it challenging for megacities across the globe to deploy UAV networks into practice. Taking Yangpu District, Shanghai, a typical central urban area, as the research object, this paper constructs a full-process optimization framework for urban UAV service networks to address the above pain points. First, building obstacles, airspace restrictions and demographic data are standardized via special gridding. Second, an improved A* algorithm is developed to generate safe and smooth 3D flight paths and compute flight duration. Third, a P-median model integrated with the Lagrangian Relaxation-Volume algorithm optimizes vertiport layout with the aim of minimizing population-weighted total flight time. Real world simulations validate the framework's advantages: the improved A* reduces flight time by 2%-8% across all distance ranges and deploying 15 vertiports yields a tiny optimality gap of 0.9174%, covering all local residents within a 5-minute flight radius. This paper offers a China-originated governance solution to support standardized and secure low-altitude economic development in megacities globally, serving as technical reference for metropolitan UAV network planning worldwide.

Introduction

Low-altitude economy with Unmanned Aerial Vehicles (UAVs) is essential in building Smart Cities, and optimized UAV service network layout is the key technical support to realize large-scale aerial logistics and public services[1]. Worldwide, engineers mostly adopt A*-series heuristic algorithms for 3D flight path searching[2] and P-median integer programming systems for vertiport location optimization[3]. However, most A* algorithms adopt fixed 8-direction grid expansion and oversimplified geometric distance heuristics, ignoring constraints like climbing gradient, which leads to jagged, time-consuming trajectories inconsistent with actual flight rules. And few studies build a full chain statistical processing system and lacks replicable operation for urban districts like downtown Shanghai. These unaddressed limitations form the critical research gap of this paper.

To address the above issues, this paper proposes an integrated optimization framework for UAV service networks targeting megacity central areas. For path planning, we design an improved A* algorithm that introduces heading adjustment and constructs a physics-based cost function. And with flight time derived from the improved A* algorithm, we implement the P-median model and adopt the Lagrangian Relaxation-Volume algorithm embedded with Teitz-Bart local search to solve large-scale location instances[4].



Background

Two fundamental methodologies dominate the research. The A* algorithm sorts candidate nodes by aggregating the actual travel cost from the starting point to the current node and the estimated cost from the current node to the destination, then iteratively expands the node with the minimum aggregated cost to guarantee both computational efficiency and global path optimality. Its core evaluation function is defined as:

$$f(n) = g(n) + h(n)$$

where $f(n)$ denotes the total comprehensive cost of node n ; $g(n)$ represents the actual flight cost from the starting grid to node n ; $h(n)$ is the heuristic estimated cost from node n to the target destination. Two mainstream heuristic functions for 3D urban flight scenarios are listed as follows.

Euclidean distance:

$$h(n) = \sqrt{(x'^2 - x^2) + (y'^2 - y^2) + (z'^2 - z^2)}$$

Manhattan distance:

$$h(n) = |x' - x| + |y' - y| + |z' - z|$$

Standard A* only generates candidate nodes along fixed directions, which restricts continuous heading adjustment for UAVs. And the heuristic functions above ignores the actual flight kinetics of UAVs, where drones must perform segmented climbing to bypass high-rise buildings.

Another important algorithm is the P-median problem. The core objective of urban UAV service network and the P-Median problem is to select P candidates as vertiports

from all available locations, such that the total population-weighted flight time from every residential demand grid to its assigned nearest vertiport is minimized. Let w_i denote the resident population weight of demand grid i , and t_{ij} represent the actual UAV flight time between grid i and candidate vertiport grid j .

Two sets of binary decision variables are defined as follows.

$y_j \in \{0,1\}$: Equals 1 if grid j is designated as a UAV vertiport; otherwise equals 0;

$x_{ij} \in \{0,1\}$: Equals 1 if demand grid i is allocated to vertiport j ; otherwise equals 0.

Therefore, the model for the population-weighted p-median problem is formulated as follows.

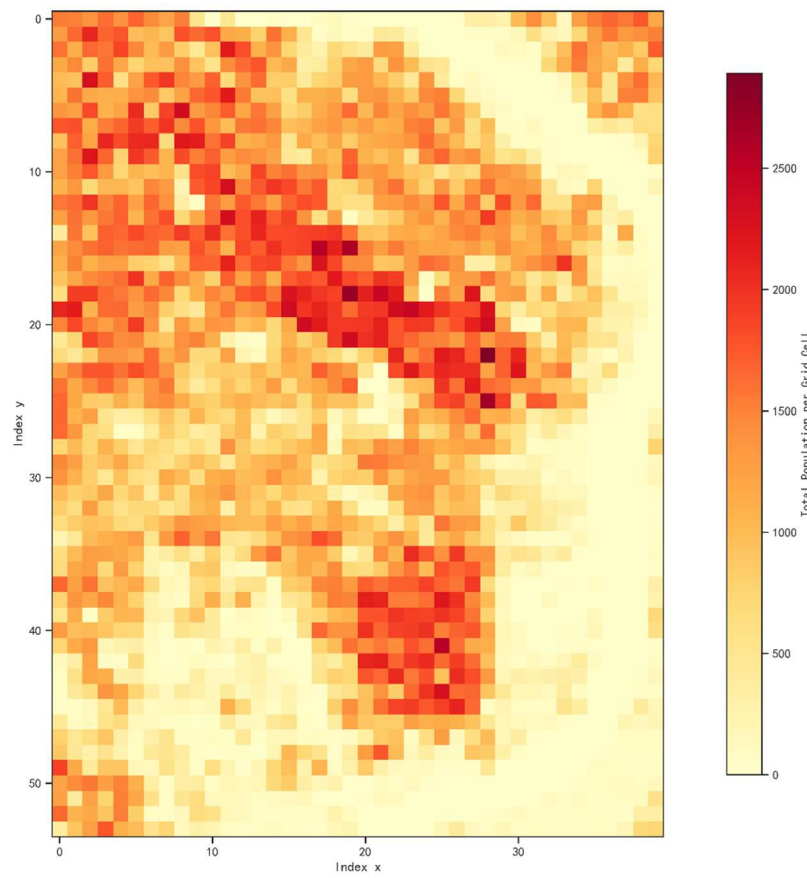
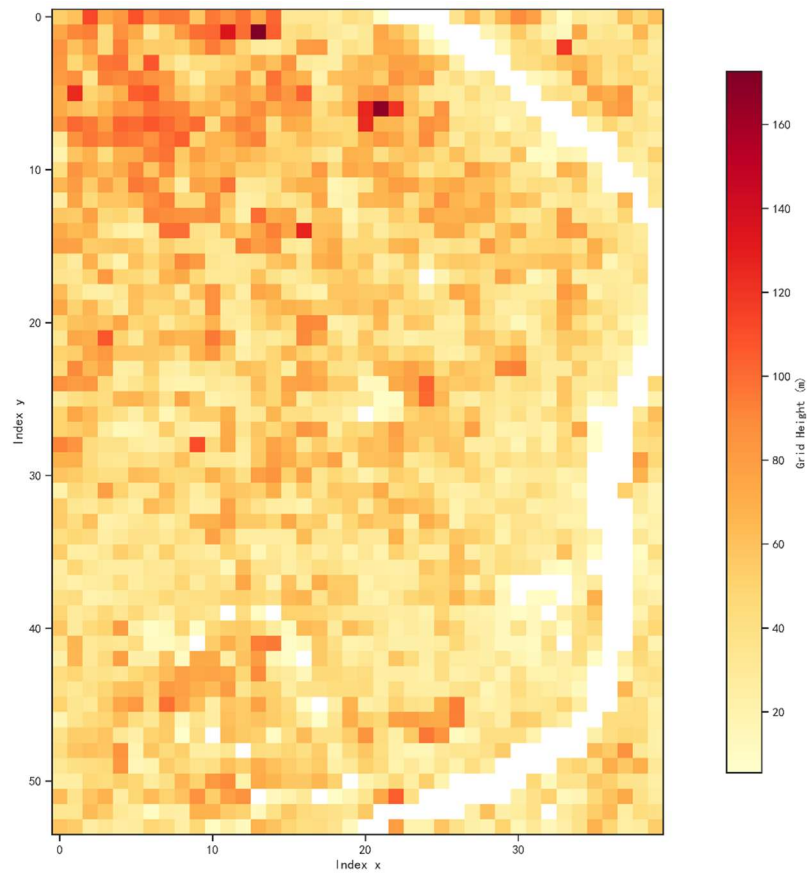
$$\begin{aligned}
\min Z &= \sum_{i=1}^N \sum_{j=1}^N w_i t_{ij} x_{ij} \\
s. t. \quad &\sum_{j=1}^N x_{ij} = 1, \forall i \in \{1, \dots, N\} \\
&x_{ij} \leq y_j, \forall i, j \in \{1, \dots, N\} \\
&\sum_{j=1}^N y_j = P \\
&x_{ij}, y_j \in \{0,1\}, \forall i, j
\end{aligned}$$

In this formulation, the objective function Z quantifies the total population-weighted flight time across all residents, measured in seconds. Constraint 1 guarantees every residential demand grid is assigned to exactly one vertiport with no uncovered demand. Constraint 2 enforces that demand grids can only be allocated to cells selected as operational vertiports. Constraint 3 restricts the total number of constructed vertiports to a fixed quantity P . Constraint 4 imposes binary integrality requirements on all allocation and site-selection decision variables.

Direct solving of this problem for large-scale grid instances brings unbearable computational overhead. Thus, Lagrangian Relaxation and Volume iterative algorithms are introduced to derive tight upper and lower bounds for near-optimal solutions.

Data Preprocessing

Urban high-rise clusters may force UAVs to frequently adjust altitude or take detours to avoid obstacles. Therefore, this study adopts spatial gridding to avoid such problem and simplify the modeling process. The research scope, Yangpu District, is discretized into $200\text{m} \times 200\text{m}$ grid units. And building height data are aggregated into each grid, with the maximum building height recorded to represent the grid. Meanwhile, population data are distributed to the corresponding grids, reflecting the service demand intensity of each unit. Additionally, all restricted zones are marked by assigning an impassable height value to matched grids, blocking illegal flight routes during path searching. Below are two heatmaps visualizing Yangpu District's grid-wise building elevation and special population.



Path Planning

Definition. After rasterizing the research scope into standardized $200\text{m} \times 200\text{m}$ grid units, we transformed UAV trajectory generation into a constrained time-minimization search problem. The problem is now searching for the cost-minimum path between corresponding grid units.

Constraints. Two types of practical constraints are integrated into the modeling framework, including UAV flight performance limits and low-altitude operation policies. All constraint indicators and their quantified standards are listed below.

Category	Indicator	Criterion
Flight Performance	Maximum speed	Speed ≤ 15.5 m/s
	Maximum vertical speed	Vertical speed ≤ 3 m/s
	Maximum gradient	Climbing Slope ≤ 0.3
	Maximum climbing height	Single altitude lift ≤ 25 m
Airspace Regulation	Minimum altitude	altitude $\geq 65\text{m}$
	Maximum altitude	altitude $\leq 115\text{m}$
	Safety distance	Gap above building top ≥ 12 m
	Scope restriction	Restricted within the open airspace of Yangpu District

Cost Function. To address the previously mentioned pain points, this paper reconstructed the cost function to better represent real-world flying patterns. The UAV scans terrain information grid by grid along the straight line connecting the start and target positions. If the current flight altitude exceeds the maximum building height of the forward grid, the UAV maintains its current altitude to pass through. If the current altitude is insufficient to clear upcoming obstacles, the aircraft ascends obliquely along the Euclidean straight path. This strategy minimizes altitude fluctuation and generates smooth trajectories while satisfying mandatory vertical safety clearance above buildings. And flight time calculation strictly complies with UAV physical performance limits, with two flight modes divided by the critical climbing gradient threshold of 0.3. When the climbing gradient is less than 0.3, the UAV maintains the maximum horizontal cruising speed of 15.5 m/s along the 3D straight path:

$$t_{ij} = \frac{\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}}{15.5}$$

When the climbing gradient exceeds 0.3, flight duration is constrained by the maximum vertical ascent speed of 3 m/s. The segmental travel time is determined by the larger value between horizontal transit time and vertical climbing time:

$$t_{ij} = \max\left(\frac{\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}}{11}, \frac{|z_i - z_j|}{3}\right)$$

To embed this physical rule into heuristic node expansion, we develop a subroutine to generate straight-line trajectories and compute corresponding flight durations between arbitrary grid pairs, which serves as the fundamental cost calculation and heuristic function module of the improved A* algorithm.

Algorithm 1 summarizes the complete workflow for trajectory and time estimation, providing precise, physically consistent cost inputs for subsequent path search.

Algorithm 1: Trajectory and Duration Calculation Between Two Grids

Input: Start coordinate (x, y) , target coordinate (x', y') , initial flight altitude h
Output: Waypoint sequence path, total flight time $time$, final flight altitude h

```

1 while  $|x - x'| > 0.5$  or  $|y - y'| > 0.5$ :
2   locate current grid
3   calculate boundary coordinate  $(next_x, next_y)$ 
4    $nexth = \max(h, GridHeight(next_x, next_y) + SafetyBuffer)$ 
5    $dist = \sqrt{(next_x - x)^2 + (next_y - y)^2}$ 
6    $dHeight = nexth - h$ 
7    $slope = dHeight / dist$ 
8   if  $slope < 0.3$ :
9      $time += \sqrt{dist^2 + dHeight^2} / 15.5$ 
10  else:
11     $time += \max(dist/11, dHeight/3)$ 
12  update coordinate  $x = next\_x, y = next\_y$ 
13   $path.append((x, y, h))$ 
14 return path, time, h
```

Improved A* Algorithm. Based on the improved cost function, we further propose three targeted upgrades to the original A* framework: flexible direction expansion, physics-based heuristic function, and embedded hard constraints. The optimized algorithm is capable of generating smooth, time-minimal, regulation-compliant 3D flight trajectories for urban UAVs.

Conventional A* only supports 8 fixed discrete moving directions, which restricts the UAV from flying straight toward the destination and creates redundant zigzag segments. During node expansion, the proposed algorithm calculates the direct heading angle pointing from the current grid to the target and appends this straight-line directional vector into the candidate direction set. Continuous step lengths along the direct path are generated dynamically, allowing UAVs to travel along the optimal geometric line between two points. This upgrade transforms discrete grid hopping into continuous spatial straight-line search. It drastically cuts redundant turns and repeated altitude adjustments, yielding smoother trajectories that match real UAV manual flight habits.

The improved evaluation metric replaces abstract geometric distance with real flight time calculated by Algorithm 1. The revised cost function retains the standard A* structure while redefining each term with physical flight duration:

$$f(n) = g(n) + h(n)$$

where $g(n)$ denotes the cumulative actual flight time from the start point to node n , and $h(n)$ represents the estimated straight-line flight time from node n to the destination.

All low-altitude regulatory and performance limits are embedded into the node expansion logic for real-time judgment and invalid branch pruning. When Algorithm 1 computes the required altitude of a candidate direction, the algorithm discards this path immediately if the predicted flight height exceeds the 115 m ceiling or the single-segment climbing height surpasses the 25 m threshold. This mechanism ensures all output trajectories fully satisfy flight regulations and can be deployed directly in real scenarios.

Built upon the priority queue structure of standard A*, the complete workflow of the improved algorithm is summarized in Algorithm 2:

Algorithm 2: Improved A* Path Planner

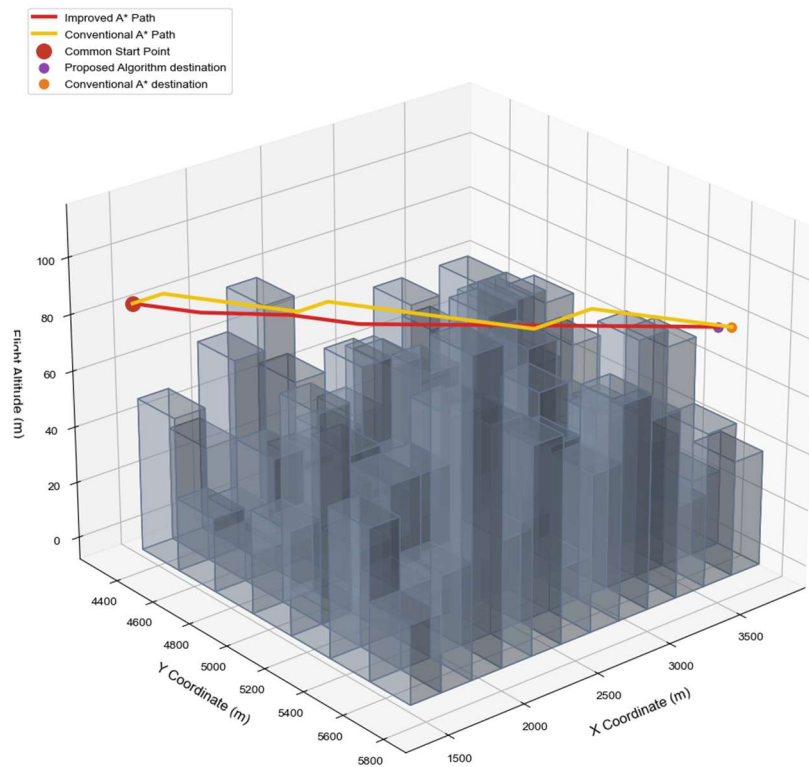
Input: Start coordinate (x, y) , target coordinate (x', y')
Output: Ordered waypoint sequence path, total flight duration time

- 1 priority queue p_heap , standard 8-direction list $dirs$
- 2 create dictionary $came_from$ for backtracking; dictionary $cost$ to record the minimal travel time to each grid node
- 3 **while** p_heap :
- 4 $time, x, y, h = \text{heapq.heappop}(p_heap)$
- 5 **if** $|x_- - x| \leq 100$ and $|y_- - y| \leq 100$:
- 6 **break**
- 7 $dir = \left(\frac{200(x_- - x)}{\sqrt{(x - x_-)^2 + (y - y_-)^2}}, \frac{200(y_- - y)}{\sqrt{(x - x_-)^2 + (y - y_-)^2}} \right)$
- 8 $dirs.append(dir)$
- 9 **for** x_{step}, y_{step} **in** $dirs$
- 10 $step_{path}, step_{time}, h_{step} = \text{Algorithm1}(x, y, x_{step}, y_{step}, h)$
- 11 **if** $h_{step} > 25$ or $h_{step} - h > 25$:
- 12 **continue**
- 13 update $time_{new}, x_{new}, y_{new}, z_{new}$
- 14 **if** $(x_{new}, y_{new}, z_{new})$ **not in** $cost$ **or** $time_{new} \leq time$:
- 15 update $came_from, cost$
- 16 $p_heap.heappush(heap, (time_{new}, x_{new}, y_{new}, z_{new}))$
- 17 backtrack path
- 18 **return** path, time

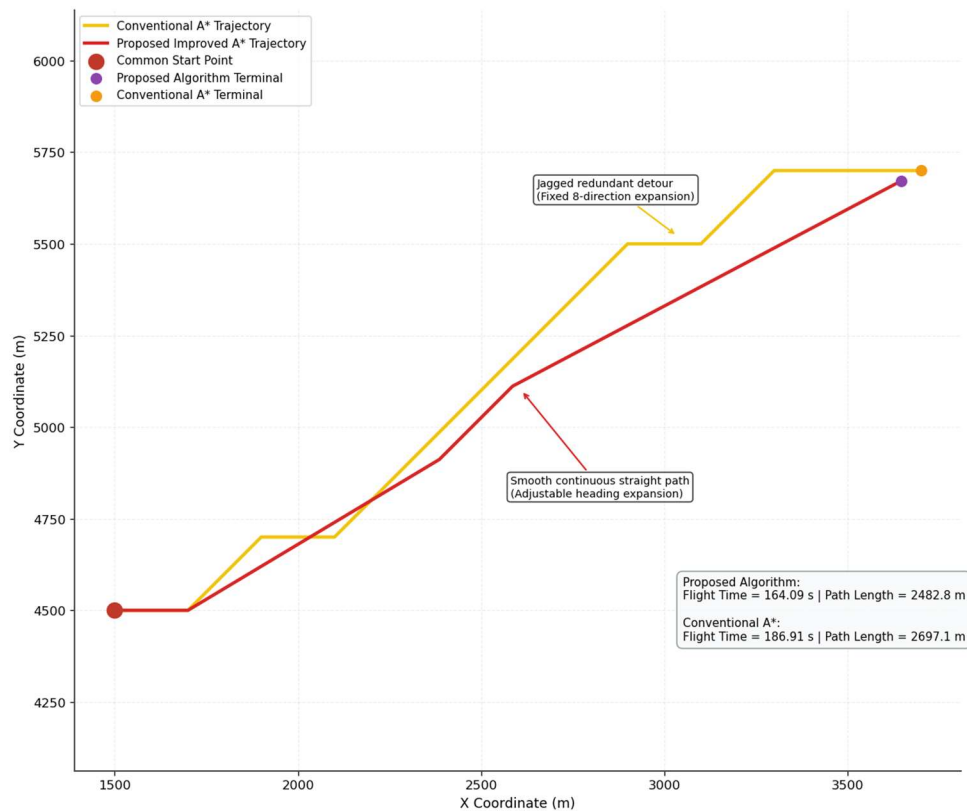
Simulation Analysis. To verify the effectiveness of the proposed algorithm, we carry out simulations against the conventional A* algorithm. All experiments are deployed on a 196×196 grid region with dense delivery demand for representative testing.

Benefiting from flexible straight-line heading expansion and terrain-adaptive gentle climbing mechanisms, trajectories generated by the proposed algorithm maintain continuous, smooth curvature and conform to real UAV flight maneuver characteristics.

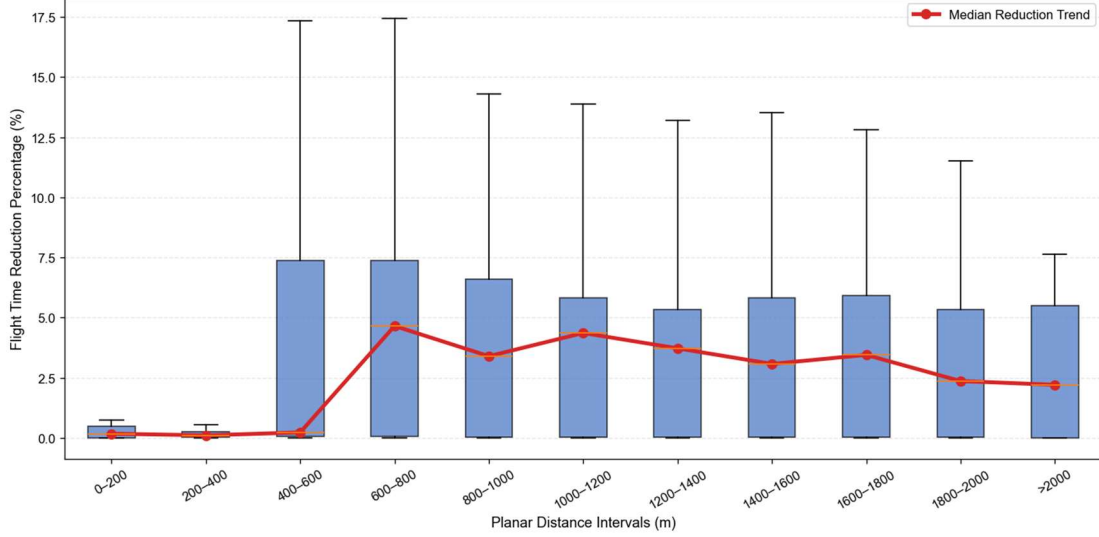
Below contrasts the 3D flight paths yielded by conventional A* and our improved algorithm within the urban obstacle environment.



And the figure below further provides their corresponding 2D top-down projection for intuitive observation of zigzag detours.



The improved A* algorithm uses adjustable directional search to eliminate redundant zigzag segments inherent to fixed-neighbor A*, which effectively cuts total consumption. Figure below illustrates the percentage reduction in flight time achieved by our algorithm across segmented travel distance ranges.



To assess the real-time performance of the improved algorithm in urban UAV navigation, both algorithms are tested under identical hardware and software environments, with statistical planning time summarized in the table below.

	Improved A*	Conventional A*
Average running time (s)	0.544433	0.480137
Median running time (s)	0.252000	0.295000

Statistical indicators reveal that the median planning time of our algorithm is lower than that of conventional A*, demonstrating faster convergence for most routine path-planning instances. The slightly smaller average runtime of the baseline A* stems from its stable but suboptimal search behavior in a small subset of complex obstacle scenarios. The traditional algorithm limits search directions to reduce computational fluctuation, yet this tradeoff introduces redundant flight segments and longer actual delivery durations.

Taking both flight time optimization gain and computational overhead into comprehensive consideration, the proposed improved A* achieves balanced overall performance and better practical adaptability for large-scale urban low-altitude UAV distribution networks.

Vertiports Deployment

Framework. As demonstrated in the Background section, the integer linear programming model of the vertiport location problem involves $N = 2160$ spatial demand grids. Direct exact solution of this integrated ILP would generate over

4.6×10^6 binary 0-1 decision variables, leading to computational overhead for large-scale urban instances. To balance computational efficiency and solution quality, this work adopts Lagrangian relaxation to decompose the tightly coupled original problem into independent subproblems. The Volume subgradient algorithm is further deployed to solve the Lagrangian dual problem. This hybrid framework simultaneously yields high-quality feasible primal solutions, providing upper bounds (UB) and rigorous theoretical lower bounds (LB), which jointly quantify the optimality gap of the obtained solutions and verify the reliability of the optimization results.

Lagrangian Relaxation Model. In the ILP problem, enforcing every demand node to be assigned to exactly one vertiport acts as the primary coupling source that increases complexity. Therefore, we relax this equality constraint into the objective function and introduce non-negative Lagrangian multipliers $\lambda_i \geq 0, (i = 1, \dots, N)$, yielding the following subproblem.

$$\begin{aligned} L(\lambda) &= \min_{x,y} \sum_{i=1}^N \sum_{j=1}^N w_i t_{ij} x_{ij} + \sum_{i=1}^N \lambda_i \left(1 - \sum_{j=1}^N x_{ij} \right) \\ &= \sum_{i=1}^N \lambda_i + \min_{x,y} \sum_{i=1}^N \sum_{j=1}^N (w_i t_{ij} - \lambda_i) x_{ij} \\ s. t. \quad &x_{ij} \leq y_j, \forall i, j \in \{1, \dots, N\} \\ &\sum_{j=1}^N y_j = P \\ &x_{ij}, y_j \in \{0, 1\}, \forall i, j \end{aligned}$$

Based on the weak duality theorem, for any $\lambda \geq 0$, $L(\lambda)$ constitutes a valid lower bound of the optimal objective Z^* , i.e., $L(\lambda) \leq Z^*$. The Lagrangian dual problem aims to maximize this lower bound to acquire the tightest possible theoretical limit:

$$Z_{LD} = \max_{\lambda \geq 0} L(\lambda)$$

Z_{LD} represents the sharpest lower bound from Lagrangian relaxation, and solving this dual problem is critical for rigorous optimality evaluation. Since $L(\lambda)$ is concave and non-differentiable with respect to λ , we adopt the Volume algorithm [6]. For any fixed multiplier vector λ , $L(\lambda)$ can be decoupled into N independent subproblems with analytical optimal solutions. We first define the marginal contribution $c_j(\lambda)$ of candidate vertiport j as:

$$c_j(\lambda) = \sum_{i=1}^N \min\{0, w_i t_{ij} - \lambda_i\} \leq 0$$

This metric quantifies the total objective reduction brought by deploying vertiport j : a demand node i will be allocated to j only if $w_i t_{ij} - \lambda_i < 0$, which reduces the relaxed objective value. To minimize $L(\lambda)$, we select the P vertiports with the smallest marginal contribution values. Let $S(\lambda)$ denote the optimal selected facility set:

$$S(\lambda) = \arg \min_{S \subseteq \{1, \dots, N\}, |S|=P} \sum_{j \in S} c_j(\lambda)$$

For the assignment of variables x_{ij} , the optimal decision rule is: set $x_{ij} = 1$ if and

only if $w_i t_{ij} - \lambda_i < 0$ and $j \in S(\lambda)$; otherwise $x_{ij} = 0$. Therefore, the closed-form optimal value of the relaxed subproblem is:

$$L(\lambda) = \sum_{i=1}^N \lambda_i + \sum_{j \in S(\lambda)} c_j(\lambda)$$

Solving $L(\lambda)$ only requires computing all $c_j(\lambda)$ and sorting the candidate facilities, with overall time complexity $O(N^2 + N \log N)$

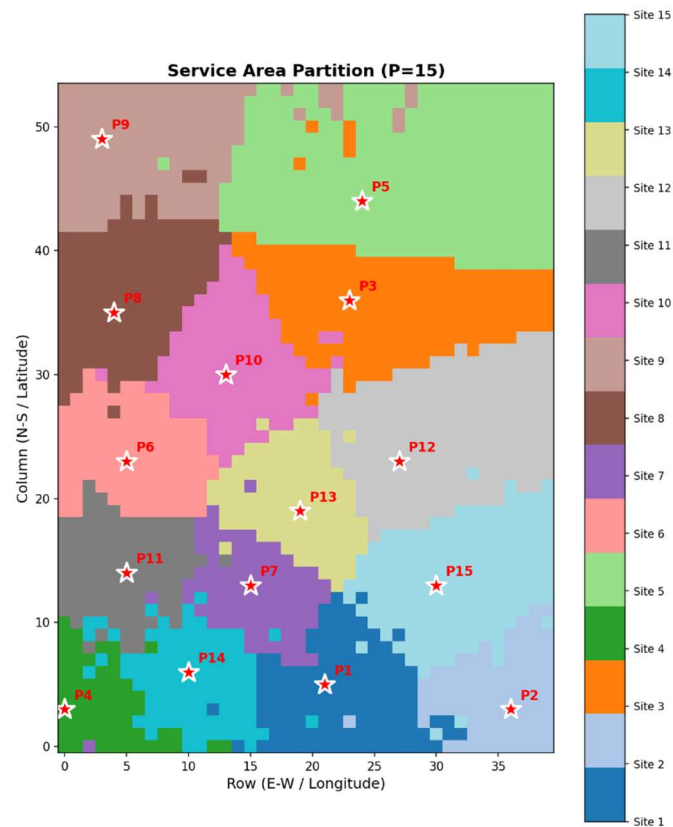
Volume Algorithm. To solve the non-differentiable Lagrangian problem and tighten the optimality gap. The Volume Subgradient Algorithm is adopted iteratively. In each iteration, $\lambda^{(k)}$ is used to update the best lower bound, while the Teitz-Bart local search refines the relaxed facility set into a fully feasible primal solution to update the best upper bound. To mitigate the oscillation and slow convergence of conventional subgradient methods, exponential smoothing is applied to stabilize the searching direction. $\lambda^{(k)}$ are then adaptively updated with a gap-dependent step size to continuously elevate the dual lower bound and shrink the duality gap. Furthermore, adaptive parameter decay and restart strategies are embedded to escape local optima when the bound improvement stagnates. The whole iterative procedure terminates when the duality gap is sufficiently small or the maximum computational budget is reached, finally yielding high-quality near-optimal vertiport deployment solutions.

Teitz-Bart Search. The Teitz-Bart heuristic is adopted for local solution improvement both inside each Volume iteration and in post-processing of the final dual solution. In each iteration, the heuristic attempts to swap one currently opened facility j_{out} with an unselected candidate node j_{in} . If the swap reduces the total weighted travel objective Z , the exchange is accepted and the facility set is updated. The search repeats exhaustively until no cost-reducing swap can be found. The computational complexity of one full pass evaluating all candidate swaps is $O(P \cdot N^2)$.

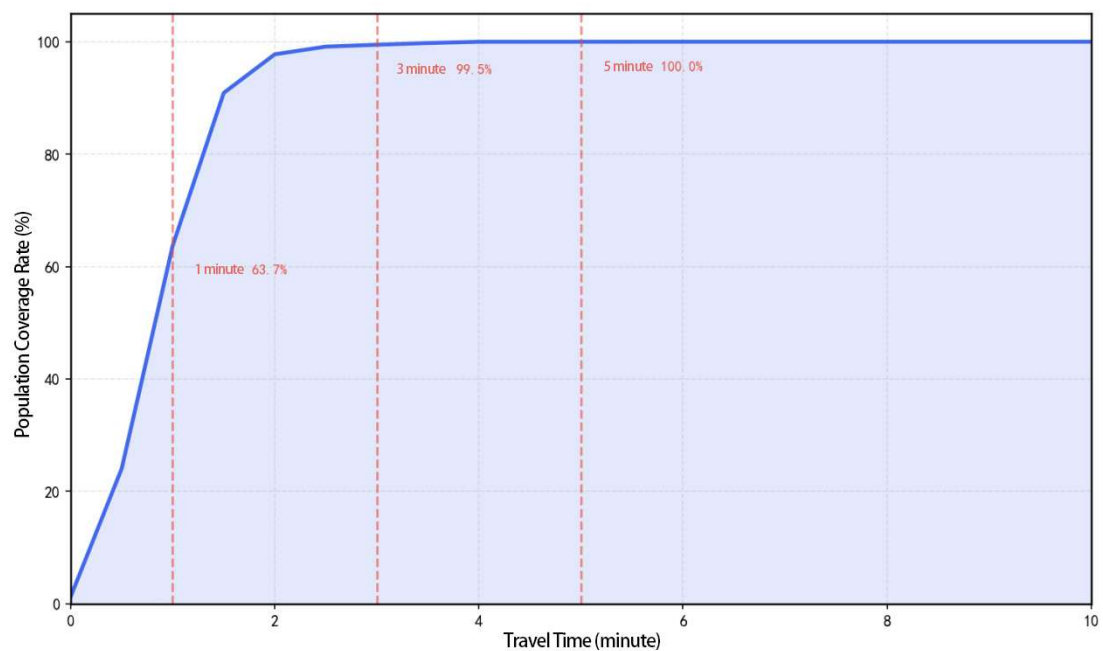
Simulation Analysis. A series of numerical experiments are conducted based on real demographics and special data from Yangpu District, Shanghai. We fixed the number of vertiports to $P = 15$ and run the Lagrangian relaxation algorithm with a 600-second time limit. The algorithm terminates after 15,901 iterations within a total runtime of 610.3 seconds. The optimal primal upper bound is $UB^* = 96154712.49$ seconds, and the tightest theoretical dual lower bound is $LB^* = 95272584.16$ seconds. The relative duality gap is calculated as:

$$\text{Gap} = \frac{UB^* - LB^*}{UB^*} \times 100\% = 0.9174\%,$$

which confirms that the obtained solution is near-globally optimal.



Each facility serves an average population of approximately 119,000 residents, covering between 62 and 343 spatial demand grids. The average flight travel time from any grid to its assigned vertiport ranges from 51.8 s to 110.6 s; most residential areas can reach the nearest vertiport within 1–1.5 minutes, indicating satisfactory overall service accessibility.



Conclusion

The 15th Five-Year Plan for National Economic and Social Development designates the low-altitude economy as a strategic emerging pillar industry. And internationally, leading economies including the EU and the United States have prioritized UAV network construction in their national infrastructure layout.

By proposing path planning and vertiports deploying algorithms, this paper demonstrates an integrated and optimized UAVs service network, which tackles the key bottleneck hindering large-scale commercial rollout of urban air mobility. Taking downtown Shanghai as the real-world research scope, this paper offers a distinctive, replicable Chinese solution for global low-altitude transportation network construction.